

Complex Number 9

1. In each case sketch the intersection of the given pair of regions.

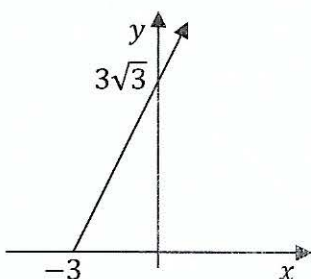
- $|z - 2 + i| \leq 2, \operatorname{Im}(z) \geq 0$
- $0 \leq \operatorname{Re}(z) \leq 2, |z - 1 + i| \leq 2$
- $|z - \bar{z}| < 2, |z - 1| \geq 1$
- $\operatorname{Re}(z) \leq 4, |z - 4 + 5i| \leq 3$
- $|z - 1 - i| \leq 2, 0 \leq \arg(z - 1 - i) \leq \frac{\pi}{4}$
- $|z| \leq 1, 0 \leq \arg(z + 1) \leq \frac{\pi}{4}$
- $|z + 1 - 2i| \leq 3, -\frac{\pi}{3} \leq \arg z \leq \frac{\pi}{4}$
- $|z - 3 - i| \leq 5, |z + 1| \leq |z - 1|$

2. A complex number z satisfies $\operatorname{Arg}(z) = \frac{\pi}{3}$

- Use a diagram to show that $|z - 2i| \geq 1$
- For which value of z is $|z - 2i| = 1$?

3. The diagram show the locus of a variable point P which represents the complex number z .

- Write down an equation for this locus
- Find the modulus and argument of z at the point where $|z|$ takes its minimum value.
- Hence find z in Cartesian form when $|z|$ takes its least value.



4. a. A complex number z satisfies $|z - 1| = 2$. Draw a diagram and hence find the greatest and least possible values of $|z|$.

b. If z is a complex number such that $\operatorname{Re}(z) \leq 2$ and $|z - 3| = 2$, show with the aid of a diagram that $1 \leq |z| \leq \sqrt{7}$

5. a. The variable complex number z satisfies $|z - 2 - i| = 1$. Use a diagram to find the maximum and minimum values of:

- $|z|$
- $|z - 3i|$.

b. A complex number z satisfies $|z| = 3$. Use a sketch to find the greatest and least values of $|z + 5 - i|$

c. The variable complex number z satisfies $|z - z_0| = r$. Use a similar approach to parts a. and b. to find the maximum and minimum values of:

- $|z|$
- $|z - z_1|$

d. Confirm your answer to the previous parts by using the triangle inequality.

6. If $x = a + b, y = a\omega + b\omega^2, z = a\omega^2 + b\omega$, where $1, \omega, \omega^2$ are the cube root of unity, prove that

- $x + y + z = 0$
- $(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega) = a^2 + b^2 + c^2 - ab - bc - ca$

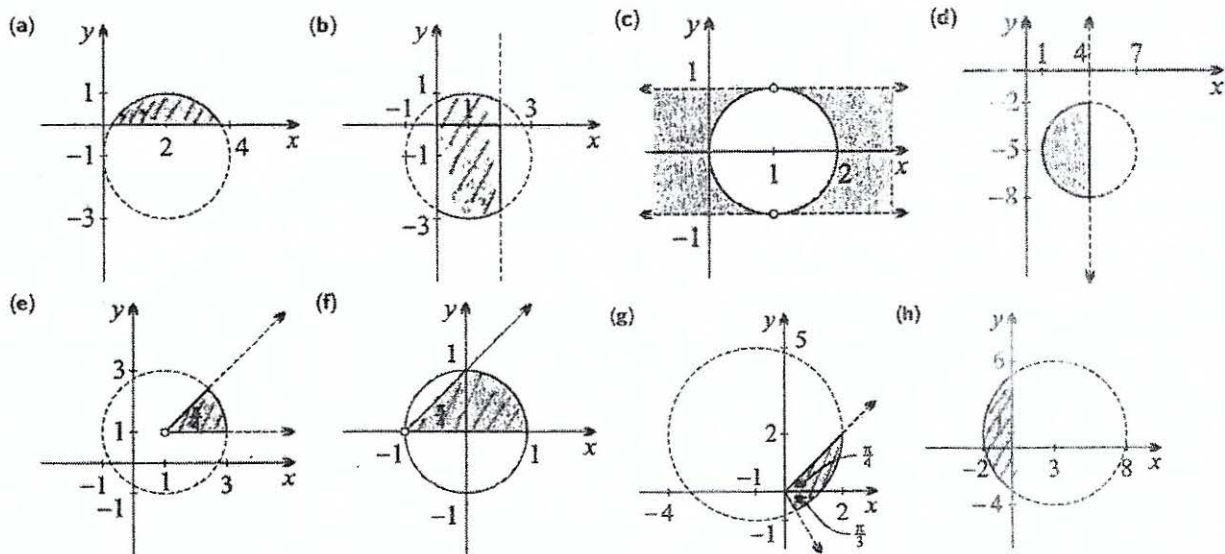
7. If $1, \omega, \omega^2$ are the three cube roots of unity, prove that $(a + b + c)(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega) = a^3 + b^3 + c^3 - 3abc$.

8. Graph the intersection of the regions defined by: $z\bar{z} \geq 9, z + \bar{z} \leq 8$ and $0 < \operatorname{Arg} z < \frac{\pi}{4}$.

9. Let α be the complex root of $z^7 = 1$ with the smallest positive argument. Let $\vartheta = \alpha + \alpha^2 + \alpha^4$ and $\delta = \alpha^3 + \alpha^5 + \alpha^6$.
- Explain why $1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 = 0$
 - Show $\vartheta + \delta = -1$ and $\vartheta\delta = 2$.
 - Write a quadratic equation in the variable t with roots in ϑ and δ .
 - Solve the quadratic equation and deduce that $\vartheta = -\frac{1}{2} + i\frac{\sqrt{7}}{2}$ and not $-\frac{1}{2} - i\frac{\sqrt{7}}{2}$.
 - Find the value of $\cos\frac{4\pi}{7} + \cos\frac{2\pi}{7} - \cos\frac{\pi}{7}$.
 - Deduce the value of $\sin\frac{4\pi}{7} + \sin\frac{2\pi}{7} - \sin\frac{\pi}{7}$.

Answer:

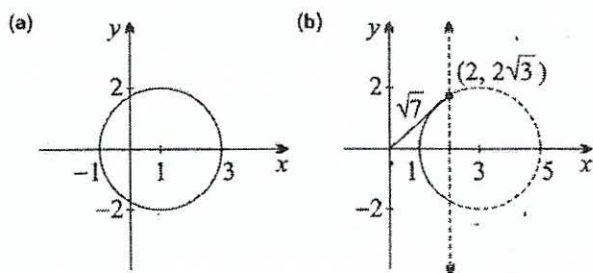
1.



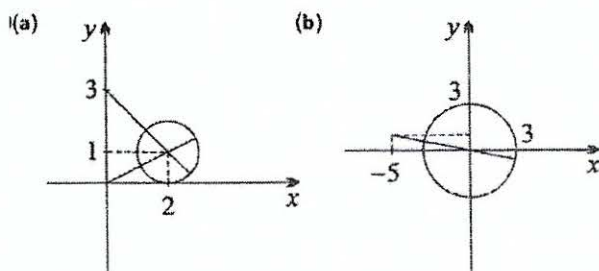
2. b. $\sqrt{3}cis\frac{\pi}{3}$

3. a. $\arg(z+3) = \frac{\pi}{3}$ b. $|z| = \frac{3\sqrt{3}}{2}, \arg z = \frac{5\pi}{6}$

4. a. 3 and 1

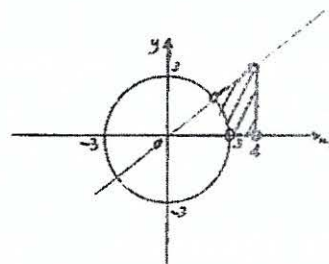


5. a. i. $\sqrt{5} + 1$ and $\sqrt{5} - 1$ $\sqrt{26} + 3$ and $\sqrt{26} - 3$
ii. $2\sqrt{2} + 1$ and $2\sqrt{2} - 1$



c. i. $||z_0| - r| \leq |z| \leq |z_0| + r$ ii. $||z_0 - z_1| - r| \leq |z - z_1| \leq |z_0 - z_1| + r$

8.



9. c. $t^2 + t + 2 = 0$ d. $t = -\frac{1}{2} \pm \frac{\sqrt{7}}{2}i, \operatorname{Im}(\theta) > 0$ e. $-\frac{1}{2}$ f. $\frac{\sqrt{7}}{2}$